Neural Importance Sampling for GW Inference: A Critical Study

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Abstract

We present a critical study of Dingo-IS [4], a method that combines amortized neural posterior estimation (NPE) with importance sampling (IS) for fast and accurate gravitational-wave parameter inference. We explain the theoretical foundations, highlight strengths and weaknesses, and reproduce the core results on controlled 2D and 3D toy posteriors. Our main original contribution is an investigation of the narrow-parameter problem, a fundamental tension in the NPE plus IS pipeline that we identify and characterize theoretically. We show that standard fixes such as alpha-divergence training and iterative IS refinement fail to resolve it, while an annealed training schedule provides a partial but principled improvement.

1 Background

Bayesian inference for gravitational waves. When a gravitational-wave (GW) detector records data d, the goal is to infer the physical parameters θ of the source, such as component masses and sky position, through the Bayesian posterior:

$$p(\theta \mid d) = \frac{p(d \mid \theta)\, p(\theta)}{p(d)}, \tag{1}$$

where $p(\theta)$ is the prior, $p(d \mid \theta)$ is the likelihood, and $p(d) = \int p(d \mid \theta) p(\theta)\, d\theta$ is the Bayesian evidence. For GW signals, the likelihood has a known closed form under a Gaussian noise model [13], but each evaluation requires a costly waveform simulation, making standard samplers such as MCMC [9, 7] or nested sampling [12] slow: a single event can take days or weeks with accurate waveform models.

Amortized neural posterior estimation. NPE [11, 1] trains a conditional normalizing flow $q_\phi(\theta \mid d)$ to approximate $p(\theta \mid d)$ using simulated training pairs (θ, d) with $\theta \sim p(\theta)$, $d \sim p(d \mid \theta)$. The training objective is the forward KL divergence:

$$\mathcal{L}_{\text{fKL}} = D_{\text{KL}}(p(\theta \mid d) \parallel q_\phi(\theta \mid d)) = -\mathbb{E}_{\theta \sim p(\cdot \mid d)}[\log q_\phi(\theta \mid d)] + \text{const.} \tag{2}$$

Once trained, q_ϕ generates posterior samples for any new event in seconds, amortizing the cost over all future events. A critical property (*mass-covering*) of $\mathcal{L}_{\text{fKL}}$ is that it diverges to $+\infty$ whenever q_ϕ assigns zero probability to a region where p has mass, which forces $\text{supp}(p) \subseteq \text{supp}(q_\phi)$.

Importance sampling and sample efficiency. Given n proposal samples $\theta_i \sim q_\phi(\theta \mid d)$, IS assigns each sample an unnormalized weight:

$$w_i = \frac{p(d \mid \theta_i)\, p(\theta_i)}{q_\phi(\theta_i \mid d)}. \tag{3}$$

The normalized weights $\bar{w}_i = w_i / \sum_j w_j$ define a corrected posterior estimate. The *effective sample size* $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$ measures how many i.i.d. samples from the true posterior the weighted set is worth. The *sample efficiency* $\epsilon = n_{\text{eff}}/n \in (0, 1]$ is the key diagnostic: $\epsilon = 1$ means the proposal is perfect, while $\epsilon \to 0$ means almost all weight falls on a handful of samples and IS is unreliable. The Bayesian evidence is estimated as $\hat{p}(d) = \frac{1}{n} \sum_i w_i$, with log-evidence uncertainty $\sigma_{\log p(d)} = \sqrt{(1-\epsilon)/(n\epsilon)}$.

2 The Dingo-IS Method

Architecture. Dingo [3] trains a conditional normalizing flow on pairs (θ, d) simulated from the GW forward model. The data d consists of frequency-domain strain in $[20, 1024]$ Hz, compressed into summary statistics via a neural network. The flow itself is based on neural spline transforms [6]. A key ingredient is *group-equivariant NPE* (GNPE) [2], which integrates known physical symmetries of the GW signal through an iterative Gibbs-sampling procedure over proxy parameters, improving accuracy at the cost of a non-tractable density.

IS correction pipeline. The Dingo-IS pipeline [4] proceeds as follows.

1. Generate $n = 10^5$ proposal samples $\theta_i \sim q_\phi(\cdot \mid d)$ using Dingo (typically under one minute on a GPU).
2. Evaluate the exact likelihood $p(d \mid \theta_i)$ for each sample in parallel across CPUs.
3. Compute IS weights via Equation 3 and derive the corrected posterior, evidence estimate, and sample efficiency ϵ.

This corrects any inaccuracies in the neural proposal while preserving full parallelizability, unlike MCMC which is inherently sequential. The median sample efficiency across 42 binary black hole events is $\epsilon \approx 10\%$, 2 orders of magnitude better than standard samplers! The log-evidence uncertainty is also improved (reduced) by a factor of 10.

The phase parameter and the grid trick. The orbital phase ϕ_c is a particular challenge: its conditional posterior $p(\phi_c \mid d, \tilde{\theta})$ given the remaining 14 parameters is nearly deterministic (very narrow) yet varies nonlinearly with $\tilde{\theta}$. Dingo, trained with forward KL, tends to over-broaden this conditional, severely reducing ϵ. The authors resolve this by *analytically marginalizing* ϕ_c from the flow and sampling it synthetically. The waveform decomposes as:

$$\tilde{h}_+(f) = \sum_m \tilde{h}_{+,m}(f)\, e^{-im\phi_c}, \tag{4}$$

so the phase dependence enters only through cheap complex exponentials after caching the mode combinations $\tilde{h}_{+,m}$. The likelihood is evaluated on a 5001-point ϕ_c grid at the cost of a single waveform simulation, and ϕ_c is sampled from the resulting empirical conditional. This makes IS weights independent of the narrow conditional entirely.

Density recovery. Because GNPE does not expose a tractable density, an auxiliary unconditional normalizing flow is trained on the GNPE proxy parameters to recover it. IS is then performed in the joint parameter space.

3 Strengths and Weaknesses

Strengths.

- **Self-certifying diagnostic.** The sample efficiency ϵ is an objective quality metric. Low ϵ automatically flags out-of-distribution events (e.g., adversarial examples) without human inspection.
- **Parallelizability.** Dingo samples are i.i.d., enabling trivial parallelization of all likelihood evaluations, unlike MCMC which is sequential.

- **Mass-coverage guarantee.** The forward KL loss ensures $\text{supp}(p) \subseteq \text{supp}(q)$, preventing biased evidence estimates from missed posterior modes.
- **Amortization.** The same trained network handles all future events in seconds, while MCMC restarts from scratch for each event.
- **Evidence precision.** IS provides an unbiased evidence estimate with $10\times$ lower uncertainty than nested sampling.

Weaknesses.

- **Grid trick requires analytical tractability.** The ϕ_c solution is specific: it relies on the fact that the phase enters the waveform analytically via Equation 4. It doesn't generalize to other narrow parameters or to posteriors where the conditional has no closed form.
- **Density recovery chain.** For GNPE, the auxiliary flow trained on proxy parameters introduces an additional approximation whose error is not quantified separately.
- **Forward KL and narrow parameters: a fundamental tension.** The mass-covering property of the forward KL forces the flow to over-broaden any conditional it cannot represent sharply. For a conditional $\mathcal{N}(\mu(\theta_1), \sigma^2)$ with small σ, the flow approximates it with width $\sigma_q \gg \sigma$, and IS efficiency scales as $\epsilon \sim \exp(-D_{\text{KL}}(p\|q))$*, which decays exponentially with the broadening ratio. The grid trick is not a general fix but a problem-specific workaround/engineering for this tension.

$\tilde{w} = p/q$. One can show that $\epsilon = 1/(1 + \chi^2(p\|q))$, where $\chi^2(p\|q) = \mathbb{E}_q[(p/q)^2] - 1$ is the chi-squared divergence. Since $\chi^2(p\|q) \geq e^{D_{\text{KL}}(p\|q)} - 1$, we obtain the bound $\boxed{\epsilon \leq \exp(-D_{\text{KL}}(p\|q))}$*.

4 Our Contribution

4.1 Reproduction on Toy Posteriors

To isolate and study the narrow-parameter problem in a controlled setting, we implement the full Dingo-IS pipeline on two toy posteriors where ground truth is available.

2D posterior. We define:

$$\theta_1 \sim \mathcal{N}(0, 1), \qquad \theta_2 \mid \theta_1 \sim \mathcal{N}(\sin(2\theta_1),\, \sigma^2), \tag{5}$$

where σ controls the narrowness of the second dimension. The nonlinear mean $\sin(2\theta_1)$ creates a curved posterior that a Gaussian proposal cannot represent, making this a meaningful test case.

3D posterior. We extend to three dimensions:

$$\theta_1 \sim \mathcal{N}(0, 1), \quad \theta_2 \mid \theta_1 \sim \mathcal{N}(0.5\,\theta_1,\ 0.8^2), \quad \theta_3 \mid \theta_1, \theta_2 \sim \mathcal{N}(\sin(\theta_1)\cos(\theta_2),\ \sigma^2), \tag{6}$$

where θ_3 mimics ϕ_c: nearly deterministic given the broad parameters but with a nonlinear mean.

Flow architecture. We implement a conditional Real-NVP [5] with affine coupling layers and alternating masks. The coupling networks are two-hidden-layer MLPs with SiLU activations. We use 8 coupling layers and hidden dimension 128 for 2D, and 10 layers for 3D. All experiments run on a single NVIDIA T4 GPU.

Faithful synthetic marginalization. In our re-implementation we tried to come up with a *faithful* version of the Dingo grid trick that treats $p(\theta_2 \mid \theta_1)$ as a black box. Rather than passing the true σ and $\sin(2\theta_1)$ directly (which would be cheating), we evaluate $\log p(\theta_1, \theta_2)$ on a 501-point grid over θ_2 for each sampled θ_1, normalize to obtain the empirical conditional, and fit a Gaussian by computing grid moments. The fitted mean and variance replace the analytical formula. This mirrors exactly what Dingo does for ϕ_c via waveform mode caching.

Results. Figure 1 shows IS efficiency as a function of σ for all methods on the 2D posterior. The faithful *synthetic* method achieves almost perfect efficiency across all values of σ, confirming the theoretical prediction: when the narrow conditional is sampled exactly, it cancels from the IS weights and efficiency is determined only by how well the 1D flow models $\theta_1 \sim \mathcal{N}(0, 1)$, a very easy task. *Forward KL* degrades monotonically from $\epsilon = 0.98$ at $\sigma = 0.5$ to $\epsilon = 0.30$ at $\sigma = 0.01$, consistent with the over-broadening tendancy described in Section 4.2. Even *reverse KL* and *0.5-divergence* fail at narrow dimensions. Our Results on the 3D posterior (Fig 2) show the same qualitative pattern, with the synthetic trick remaining stable across all σ values. These observations are coherent with the paper's reported leading efficiency of $\approx 10\%$ for the full 15-dimensional GW posterior.

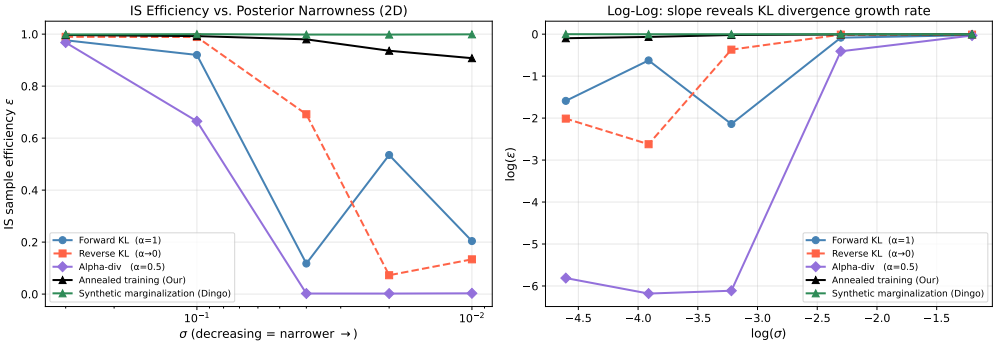

Figure 1: IS sample efficiency and KL growth rate on the 2D toy posterior (Eq 5)

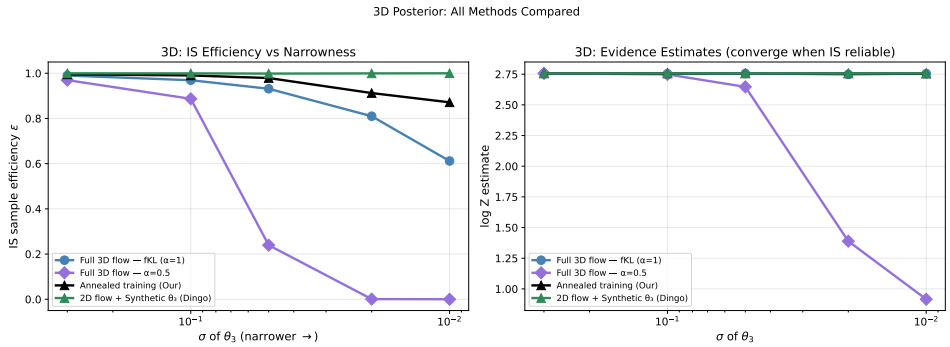

Figure 2: IS sample efficiency and log evidence estimate on the 3D toy posterior (Eq 6)

4.2 Probing the Narrow-Parameter Tension

The forward KL and IS efficiency form a fundamental tension. The mass-covering property that makes IS unbiased is exactly what forces the flow to over-broaden narrow conditionals, reducing efficiency. We characterize this tension theoretically and test several methods to resolve it.

Theoretical intuition. For a flow that approximates $\mathcal{N}(\mu(\theta_1), \sigma^2)$ with $\mathcal{N}(\mu(\theta_1), \sigma_q^2)$ where $\sigma_q = c\sigma$, $c > 1$, the KL penalty from the narrow dimension alone is:

$$D_{\mathrm{KL}}(\mathcal{N}(\mu, \sigma^2) \| \mathcal{N}(\mu, \sigma_q^2)) = \frac{1}{2}\left(\frac{1}{c^2} - 1 + 2\log c\right), \tag{7}$$

and IS efficiency decays as $\epsilon \approx \exp(-D_{\mathrm{KL}})$ (at most as $1/c$). As $\sigma \to 0$ with fixed flow capacity, $c \to \infty$ and $\epsilon \to 0$ regardless of training objective. The synthetic trick overcomes this by entirely removing the narrow dimension from the flow, so its contribution cancels in the IS weights.

Alpha-divergence training. We test the Renyi-α training objective [8]:

$$\mathcal{L}_\alpha = -\frac{1}{\alpha}\log \mathbb{E}_q\left[\left(\frac{p(\theta)}{q(\theta)}\right)^\alpha\right], \tag{8}$$

which interpolates between reverse KL ($\alpha \to 0$) and forward KL ($\alpha \to 1$). The intuition is that intermediate α might balance mass-coverage and sharpness. We test $\alpha = 0.5$. As shown in Figure 1, alpha-divergence training performs *worse* than forward KL across all values of σ.

Iterative IS refinement. We test an iterative approach: train a flow with forward KL, compute IS weights, resample proportionally to weights to obtain corrected samples, and retrain the flow on the resampled dataset. This idea is appealing but fails in practice due to *weight degeneracy*: at low ϵ, a few samples receive nearly all the weight. Resampling then produces a poor dataset (w/ massive duplication), and the retrained flow learns a bad approximation. This is quite similar to the mode collapse we observe when recursively training LLMs on synthetic data.

Annealed training. We propose training the flow on a geometric schedule of posteriors $p_t(\theta) \propto p(\theta_1) \cdot \mathcal{N}(\theta_2; \mu(\theta_1), \sigma_t^2)$ with σ_t decreasing from $\sigma_{\text{start}} = 1.0$ to σ_{target} over $T = 8$ to 10 steps, warm-starting the flow at each step. This is inspired by annealed importance sampling (AIS) [10], which uses a geometric path between distributions. Our intuition is that adjacent distributions in the schedule differ by a small $\Delta\sigma$, so the forward KL over-broadening at each step is small. Formally, if q_t over-broadens p_t by factor $c_t \approx 1 + \epsilon_t$ at step t, the final broadening is $\prod_t c_t$, which is much smaller than the single-step broadening c_T when the schedule is fine enough.

As shown in Figure 1 and Figure 2, *annealed training*, achieves clearly higher efficiency than forward KL at small σ and approaches DINGO-IS sophisticated pipeline. Annealed training reaches a better local optimum than one-shot forward KL because the flow is never asked to bridge a large distribution gap in a single training run. However, approximation errors still accumulate across steps, and the final optimum remains slightly over-broad relative to the true narrow conditional. That's why efficiency improves but does not reach $\epsilon \approx 1$.

A summary of all methods is given in Table 1.

Table 1: Comparison of methods on the narrow-parameter problem.

Method	IS unbiased	ϵ at small σ	Needs conditional	Computational cost
Forward KL	✓	Degrades	**No**	**1x** (baseline)
Reverse KL	×	Degrades	**No**	**1x**
Alpha-div ($\alpha = 0.5$)	×	Worse than fKL	**No**	**1x**
Iterative IS	✓	Degrades	**No**	4-5x
Annealed training (our)	✓	**in [0.85, 0.9]**	**No**	8-10x
Synthetic (Dingo-IS)	✓	$\approx$ **1**	Yes	**1-2x**

Conclusion. Our experiments support the following negative result: the narrow-parameter problem is difficult to be fully solved by changing the training objective alone. The synthetic trick is apparently not just an engineering convenience. Annealed training is provably useful to considerably reduce the gap at the cost of an order of magnitude in inference speed. Most importantly, it is more general than Dingo-IS: not requiring analytical conditional nor a prior knowledge of the narrow dimension(s).
We believe that preserving these benefits while being as sample efficient and as fast as Dingo-IS requires a fundamentally different inference setup.

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